

INTRADAY OPEN INTEREST ESTIMATION FRAMEWORK

Estimating Net Position Opening from the Futures Trade Stream

Purpose

This report evaluates a proposed method for estimating intraday changes in futures open interest from trade, price, footprint, and order-book data. It explains what can be estimated, what cannot be directly observed, how the proposed v0 and v1 models should be corrected, and how to build a practical production architecture.

Technical Design and Research Note

July 2026

Executive Summary

The proposed idea is a useful starting point, but its output must be described correctly. A futures trade stream does not reveal whether the buyer and seller are opening or closing positions. Therefore, the system cannot reconstruct true intraday open interest. It can only estimate the expected net change in open interest under a probabilistic model.

$$\text{Estimated OI}(t) = \text{OI anchor} + \sum q_i \times \hat{w}_i$$

Here, q_i is the trade quantity and $\hat{w}_i$ is the estimated net-opening propensity for that trade or time interval. The central research problem is estimating $\hat{w}$ from observable features.

Core conclusion

Build two separate products: (1) an end-of-day ΔOI forecast that can be trained and objectively validated, and (2) a latent intraday OI path estimate that is useful as a behavioral signal but cannot be directly verified trade by trade.

- The simple v0 model is suitable as a baseline and implementation test, but time-of-day coefficients must be estimated across many days rather than copied from a daily ratio.
- The proposed v1 daily regression is mainly an end-of-day predictor. It should not be applied directly to incomplete running features unless it is trained on intraday snapshots.
- DOM, sweep, absorption, refill, price displacement, and churn metrics can refine the estimate, but none of them proves that a trade opened or closed a position.
- Contract roll, expiration, and spread activity must be modeled explicitly because contract-level OI changes can reflect migration between expiries rather than a change in total market exposure.

1. What Open Interest Changes Mean

Every futures transaction has one buyer and one seller. The trade quantity alone does not tell us whether either participant is creating a new position or offsetting an existing position. For a transaction of quantity q , the net open-interest contribution has three possible outcomes:

Buyer action	Seller action	Change in OI
Open long	Open short	+q
Close short	Close long	-q
Open long	Close long	0
Close short	Open short	0

The aggressor side does not resolve this ambiguity. A market buy may be a new long entering, a short covering, or a combination of both. Likewise, a market sell may be new short initiation, long liquidation, or both.

$$w = P(\text{open-open}) - P(\text{close-close}), \quad \text{where } -1 \leq w \leq +1$$

$$E[\Delta OI_i] = q_i \times w_i$$

The variable w is therefore an expected contribution, not an observed label. This distinction must remain explicit throughout the software, model naming, user interface, and downstream trading logic.

2. Evaluation of the Proposed v0 Baseline

2.1 Original concept

For each historical day, compute official daily ΔOI divided by total volume. Treat the resulting signed ratio as the average net-opening fraction and allocate it across the intraday trade stream. A basic real-time estimate is:

$$\hat{OI}(t) = OI \text{ anchor} + \sum \text{bucket volume}(t) \times f(\text{bucket})$$

Operationally, this is attractive. It is a prefix sum, requires only a few counters, and can be updated in constant time per trade. It also fits naturally beside an existing cumulative-delta cache.

2.2 Strengths

- Very low implementation cost and minimal runtime overhead.
- Provides a reproducible baseline for comparing all later models.
- Creates a session-dependent prior rather than assuming that all trading hours have the same opening/closing behavior.
- Easy to reconcile against official daily OI and easy to monitor for bias.

2.3 Main statistical limitation

Daily official ΔOI does not reveal which time bucket created the change. Suppose overnight volume is 100,000 contracts, the U.S. session volume is 300,000 contracts, and the official daily ΔOI is +20,000. The available observation gives only one equation:

$$100,000 f(\text{overnight}) + 300,000 f(\text{U.S. session}) = 20,000$$

There are multiple possible pairs of coefficients that satisfy this equation. Therefore, separate time-bucket propensities cannot be identified from one day or by simply copying the daily ΔOI /volume ratio into each bucket.

2.4 Corrected v0 formulation

Estimate bucket coefficients across many days using a constrained or regularized regression:

$$\Delta OI(d) = \beta_1 V(d, \text{overnight}) + \beta_2 V(d, \text{pre-open}) + \beta_3 V(d, \text{NY-AM}) + \beta_4 V(d, \text{NY-PM}) + \varepsilon(d)$$

Each coefficient β_k becomes an empirical estimate of the average expected OI contribution per traded contract in that bucket. Ridge regularization is useful because bucket volumes are correlated. Coefficients may also be constrained to a plausible range, such as $[-1, +1]$, although the fitted values will normally be much smaller in magnitude.

2.5 Behavioral weakness of v0

Within a bucket, every contract receives the same sign and weight. If the historical coefficient is positive, all trades mechanically increase estimated OI, even during an obvious liquidation wave. The model therefore reflects average historical behavior, not the current day's changing regime.

Recommended role for v0

Use v0 as a historical prior, architecture test, and benchmark. Do not treat it as a standalone trading signal or as a reliable reconstruction of current OI.

3. Evaluation of the Proposed v1 Regression

3.1 Proposed daily features

The suggested v1 model regresses official daily ΔOI on aggregates already available in the pipeline, such as total volume, absolute net delta, price change, range, large-lot volume, small-lot volume, and trade count. A representative model is:

$$\Delta OI(d) = \beta_0 + \beta_1 \text{Volume} + \beta_2 |\text{Net Delta}| + \beta_3 \text{Price Move} + \beta_4 \text{Range} + \beta_5 \text{Large Lots} + \beta_6 \text{Small Lots} + \beta_7 \text{Trade Count} + \epsilon$$

Ridge regression is a reasonable first estimator because many of these features are strongly correlated. The model can be re-fitted whenever a new official OI observation becomes available.

3.2 What this model actually predicts

In its original form, v1 is an end-of-day ΔOI predictor. It learns relationships between completed daily features and the final daily OI change. It does not automatically learn where within the day the OI change occurred.

3.3 Why applying daily coefficients intraday is invalid

At 10:00 a.m., daily features are incomplete. The current range is not the final range, current volume is not total volume, and current price displacement from the prior close is not the final close-to-close move. Feeding running values into a model trained only on completed days changes the statistical meaning and scale of its inputs.

- A daily range of 35 points at 10:00 may finish at 110 points.
- A positive morning price move may reverse before the close.
- The proportion of large-lot volume may change materially later in the session.
- A non-zero intercept may create an artificial OI jump at the start of the day.

3.4 Corrected v1 training design

Train on historical intraday snapshots. For each day, create rows at fixed times or event-volume intervals. Every row contains features accumulated only up to that point, while the target remains the official final ΔOI for that day.

$X(d,t)$ = features observed up to time t

$y(d,t)$ = official ΔOI for day d

Include time-of-day, elapsed-session fraction, expected remaining volume, volatility regime, and contract lifecycle features. The resulting model answers: “Given what has happened so far, what is the expected end-of-day ΔOI ?”

4. Two Outputs That Must Remain Separate

4.1 End-of-day ΔOI forecast

$$\Delta \hat{OI}(\text{EOD} \mid \text{information through } t)$$

This is a supervised prediction problem. The target becomes known when the official OI update is published. Forecast accuracy can be measured with MAE, RMSE, directional accuracy, calibration by predicted magnitude, and regime-specific errors.

4.2 Current intraday OI path estimate

$$\Delta \hat{OI}(\text{current}, t) = \sum q_i \hat{w}_i, \text{ for trades } i \text{ observed through } t$$

This is a latent allocation problem. The true path is normally unavailable, so it cannot be validated at the trade level. A model may end the day near the correct official ΔOI while assigning the change to the wrong intraday periods.

Important limitation

A low end-of-day error does not prove that the estimated intraday path was correct. The model may have reached the right destination through the wrong path.

5. Review of the Proposed Features

5.1 Total volume

Necessary but weak by itself. High volume may represent new risk creation, liquidation, position transfer, short-term churn, spread execution, or roll activity. The relationship with ΔOI is conditional rather than direct.

5.2 Net delta and absolute net delta

Aggressive imbalance measures direction and intensity, not opening or closing. Absolute net delta also discards direction. Keep multiple representations: positive delta, negative delta, signed net delta, absolute net delta, and interactions with price response.

5.3 Price response

The combination of delta and price progress is more informative than either alone. Large positive delta with little upward progress suggests absorption; large negative delta with little downward progress suggests bid-side defense. However, neither pattern identifies participant inventory states with certainty.

Price behavior	Aggressive flow	Possible interpretation
----------------	-----------------	-------------------------

Rising	Positive delta	Effective aggressive buying
Flat	Positive delta	Buy-side aggression absorbed by passive sellers
Falling	Negative delta	Effective aggressive selling
Flat	Negative delta	Sell-side aggression absorbed by passive buyers

5.4 Large-lot activity

Large prints should not be hard-coded as position-initiating. They may represent new positions, institutional exits, hedging, calendar spreads, roll activity, or liquidation. Let the model learn the relationship from the archive and separate context-specific forms of large-lot activity.

- Aggressive large buy volume and aggressive large sell volume.
- Large passive-order fills and replenishment behavior.
- Large trades near versus far from short-term VWAP.
- Large trades during breakouts versus inside balance.
- Large trades followed by continuation versus immediate rejection.

5.5 Trade count and size distribution

Trade count is highly correlated with volume. More descriptive features include average and median trade size, upper size percentiles, the share of volume from large-size buckets, and sweep-event counts. Ridge regression can manage collinearity, but feature design should still reflect market behavior.

6. DOM and Footprint Feature Refinement

Order-book and footprint data can adjust the time/session prior. These signals should be interpreted as probabilistic evidence, not direct labels.

6.1 Features consistent with position initiation

- Aggressive sweeps clearing multiple price levels.
- High sweep quantity and rapid liquidity consumption.
- Trading far from short-term VWAP or outside the prior balance area.
- Breakout accompanied by price continuation rather than immediate rejection.
- Persistent one-sided aggressive activity across several events.
- Fresh resting liquidity appearing behind the direction of travel.
- Successful attack through a previously defended passive-liquidity zone.

6.2 Features consistent with closing, scratching, or churn

- High volume within a very narrow price range.
- Repeated two-way rotation with little net displacement.
- Large delta without corresponding price progress.
- Rapid alternation between aggressive buyers and sellers.

- Resting liquidity withdrawing after fills rather than replenishing.
- Repeated execution around VWAP or a high-volume node.
- Rising trade count without directional expansion.

6.3 Why these signals remain ambiguous

A sweep far from VWAP may represent new directional initiation, but it can also be forced liquidation. Absorption may indicate strategic passive entry, passive exit, hedging, or inventory transfer. The features improve probabilities only when combined with context, persistence, price response, and regime information.

7. Recommended Three-Layer Architecture

Layer 1 - Historical prior

Estimate a baseline propensity from stable contextual variables:

$$w(\text{prior}) = f(\text{time bucket, session, day of week, volatility regime, days to expiry, roll regime})$$

This layer should be simple, stable, and interpretable. The corrected time-bucket Ridge model is suitable here.

Layer 2 - Behavioral adjustment

Use current trade, footprint, price, and DOM features to move the propensity above or below the prior:

$$\hat{w}(t) = \text{clip}[w(\text{prior}, t) + g(X(t)), -1, +1]$$

$$\Delta \hat{OI}(\text{current}, t) = \Delta \hat{OI}(\text{current}, t-1) + \sum q_i \hat{w}_i$$

The adjustment model may begin as a linear or logistic transformation and later evolve into gradient boosting or a temporal model. Interpretability and calibration matter more than model complexity in the early stages.

Layer 3 - Daily reconciliation and learning

$$\text{Error}(d) = \text{official } \Delta OI(d) - \text{estimated } \Delta OI(d)$$

When official OI arrives, store the error and update model diagnostics. Refit on a rolling window or use controlled online learning. Do not blindly add yesterday's residual as an offset to today, because the residual may come from a different volatility, session, or roll regime.

- Track bias by session, weekday, volatility regime, and days to expiry.
- Monitor coefficient drift and feature-distribution drift.
- Generate a confidence score from historical residual dispersion in comparable regimes.
- Fall back to the simpler prior when live features are missing or abnormal.

8. Contract Roll and Lifecycle Effects

Futures OI must be interpreted at both the individual-contract and contract-family levels. Near rollover, OI can fall sharply in the expiring contract and rise in the next expiry while total market exposure remains nearly unchanged.

- Track front-month, next-month, and relevant deferred contracts separately.

- Identify calendar-spread and roll windows when possible.
- Include days-to-expiry and roll-regime flags in every model.
- Report both contract-level estimated OI and aggregated family-level exposure when data coverage permits.

Failure mode

Without roll controls, the model may classify migration from one expiry to another as genuine market-wide closing in the old contract and opening in the new contract.

9. Practical Implementation Roadmap

1. **Baseline 0 - Constant ratio:** Compute a rolling historical mean of daily ΔOI / daily volume. Apply one constant coefficient to all trades. Use only as a sanity check.
2. **Baseline 1 - Time-bucket Ridge:** Regress daily ΔOI on session/bucket volumes across many days. Add lifecycle and regime segmentation. This becomes the historical prior.
3. **EOD snapshot predictor:** Create intraday historical snapshots and train a model to forecast final daily ΔOI from information available at each snapshot.
4. **Behavioral intraday allocator:** Estimate event- or interval-level propensity using sweep, absorption, refill, displacement, VWAP distance, churn, and large-lot features.
5. **Reconciliation and confidence:** Compare final estimates with official ΔOI , track residuals by regime, calibrate uncertainty, and implement fallbacks.
6. **Production integration:** Expose the model through a stateful $O(1)$ -per-trade engine with stable interfaces for replay and live API data.

10. Suggested Engine Design

The estimator should be independent from other analytical engines but consume normalized trade and DOM events. The same interface should work for historical replay and live feeds.

10.1 Core state

- Official OI anchor and anchor date.
- Current estimated cumulative ΔOI .
- Per-session and per-bucket volume counters.
- Running signed and absolute delta.
- Price, VWAP, range, trade-count, and size-distribution statistics.
- Sweep, absorption, refill, and churn aggregates.
- Current contract lifecycle and roll-regime context.
- Latest EOD forecast, intraday estimate, confidence, and model version.

10.2 Event update flow

7. Receive normalized trade event and associated book context.
8. Update running feature state in $O(1)$ or amortized $O(1)$.
9. Read the historical prior for the current context.
10. Apply the behavioral adjustment model.

11. Add $q \times \hat{w}$ to the latent intraday ΔOI estimate.
12. Update the EOD ΔOI forecast from the current snapshot state.
13. Emit estimate, forecast, confidence, and diagnostics.

10.3 Example output schema

Field	Meaning
oi_anchor	Latest official OI value used as the starting point.
estimated_current_oi	Anchor plus latent cumulative estimated ΔOI .
estimated_intraday_delta_oi	Current latent change since the anchor.
forecast_eod_delta_oi	Expected official ΔOI at the end of the trading day.
net_opening_propensity	Current $\hat{w}$ value or interval-average value.
confidence_score	Calibrated reliability based on comparable historical residuals.
regime	Session, volatility, lifecycle, and roll context.
model_version	Exact model and feature definition used.
diagnostics	Top feature contributions, missing-data flags, and fallback state.

11. Validation Framework

11.1 End-of-day metrics

- Mean absolute error and root mean squared error of daily ΔOI .
- Directional accuracy: whether the sign of official ΔOI was predicted correctly.
- Error normalized by daily volume and by official OI.
- Calibration by predicted magnitude and confidence bucket.
- Performance by session, volatility regime, weekday, and days to expiry.
- Comparison against constant-ratio and time-bucket baselines.

11.2 Intraday plausibility tests

Because the true path is not observed, validate consistency rather than claiming ground-truth accuracy:

- No impossible jumps when no volume has traded.
- Bounded per-trade contribution within $[-q, +q]$.
- Stable behavior under historical replay and deterministic reprocessing.
- Reasonable response to known roll periods and abnormal sessions.

- Ablation tests showing whether DOM/footprint features improve final-day error or calibration.
- Event studies around strong liquidation, breakout, absorption, and two-sided churn episodes.

11.3 Backtesting discipline

- Use walk-forward validation rather than random train/test splits.
- Prevent leakage from final-day values into earlier snapshots.
- Use only information that would have been available at each historical timestamp.
- Separate contract-roll regimes and avoid mixing revised or future official OI values.
- Freeze feature definitions and model versions for reproducible comparisons.

12. Recommended First Production Version

The first deployable version should remain deliberately simple and transparent:

- A time-bucket Ridge prior trained on daily bucket volumes and official ΔOI .
- A small linear behavioral adjustment using signed delta, price displacement, VWAP distance, sweep depth, refill intensity, and churn.
- A separate snapshot-based Ridge model for the end-of-day ΔOI forecast.
- Daily reconciliation, rolling re-fit, regime-level error tracking, and confidence intervals.
- Explicit roll-regime logic and contract-family reporting.
- A strict naming convention that uses “estimated” and never presents the output as observed OI.

Why start simple

The labels are weak and the intraday target is latent. A complex model can fit noise and create unjustified confidence. A transparent baseline makes it possible to determine whether each added feature provides genuine out-of-sample value.

13. Final Assessment

The proposal is worth implementing. It is more directly aligned with the real question of position opening and closing than trade-classification methods such as Tick Rule or Lee-Ready. Its computational shape is also compatible with an existing cumulative stream-processing engine.

However, the system must be framed as probabilistic estimation. The v0 time-of-day model requires cross-day regression to identify bucket coefficients. The v1 daily regression must be redesigned as an intraday-snapshot model before its outputs can be interpreted during the session. DOM and footprint features can improve the estimate, but they do not reveal participant inventory states directly.

Supervised and testable: forecast of final daily ΔOI

Latent and probabilistic: estimated intraday path of OI

Keeping these two products separate will produce a system that is technically honest, measurable, and useful. The end-of-day forecast provides a verifiable learning target, while the latent intraday path provides a contextual behavioral signal for the trading engine. Confidence, regime controls, and daily reconciliation are essential parts of the design, not optional extras.

Appendix A - Minimal Mathematical Specification

For each trade i at time t :

q_i = traded quantity

X_i = observable trade, price, footprint, DOM, session, and lifecycle features

$\hat{w}_i = E[\Delta OI_i / q_i \mid X_i]$

$\Delta \hat{OI}(\text{current}, t) = \sum(i \leq t) q_i \hat{w}_i$

$\hat{OI}(\text{current}, t) = OI(\text{anchor}) + \Delta \hat{OI}(\text{current}, t)$

$\Delta \hat{OI}(\text{EOD}, t) = h(\text{running snapshot features through } t)$

$\text{Residual}(d) = \text{official } \Delta OI(d) - \Delta \hat{OI}(\text{current}, \text{end of day})$

The behavioral model should preserve the bound $-1 \leq \hat{w}_i \leq +1$. In practice, values are expected to cluster much closer to zero because a large proportion of volume transfers positions without changing total OI.

Appendix B - Key Naming Recommendations

Avoid	Prefer
Real-Time OI	Estimated Intraday OI
True Open Positions	Estimated Net Open-Position Change
OI from Trades	Trade-Stream OI Estimate
Opening Detector	Net-Opening Propensity Model
Current OI Signal	Latent Intraday OI Path